

AI-Powered Customer Churn Analytics in SaaS

Data Science • NLP • Business Intelligence

Project Story:

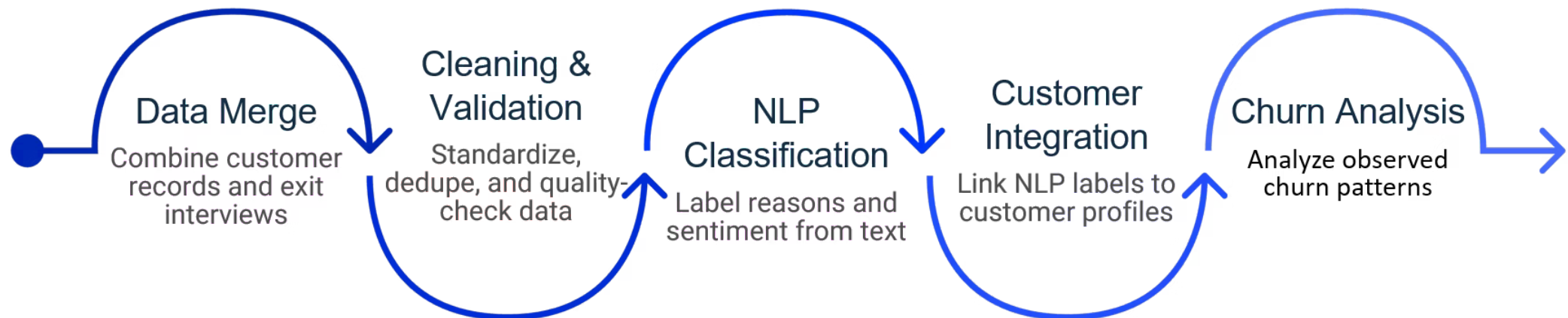
SaaS businesses generate large amounts of customer data, but understanding churn requires combining structured customer information with unstructured customer feedback. This project develops an integrated analytics framework that connects customer data, exit interviews, NLP classification, churn analysis, and interactive business intelligence.

Key Project Scale:

- 95,380 raw customer records
- 95,000 validated unique customers
- 3,405 NLP-classified exit interviews
- 3,272 matched interview records
- 3,244 unique customers represented in matched interviews

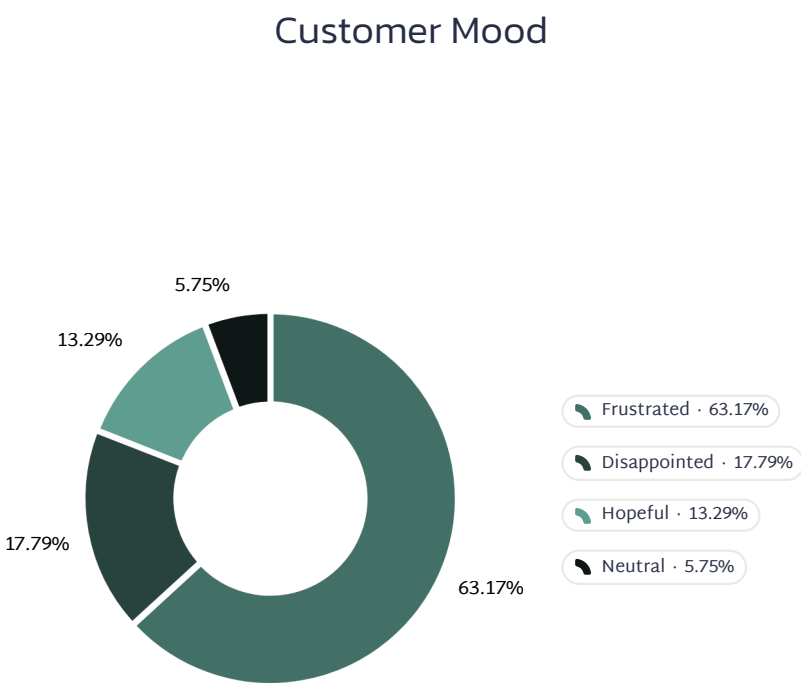
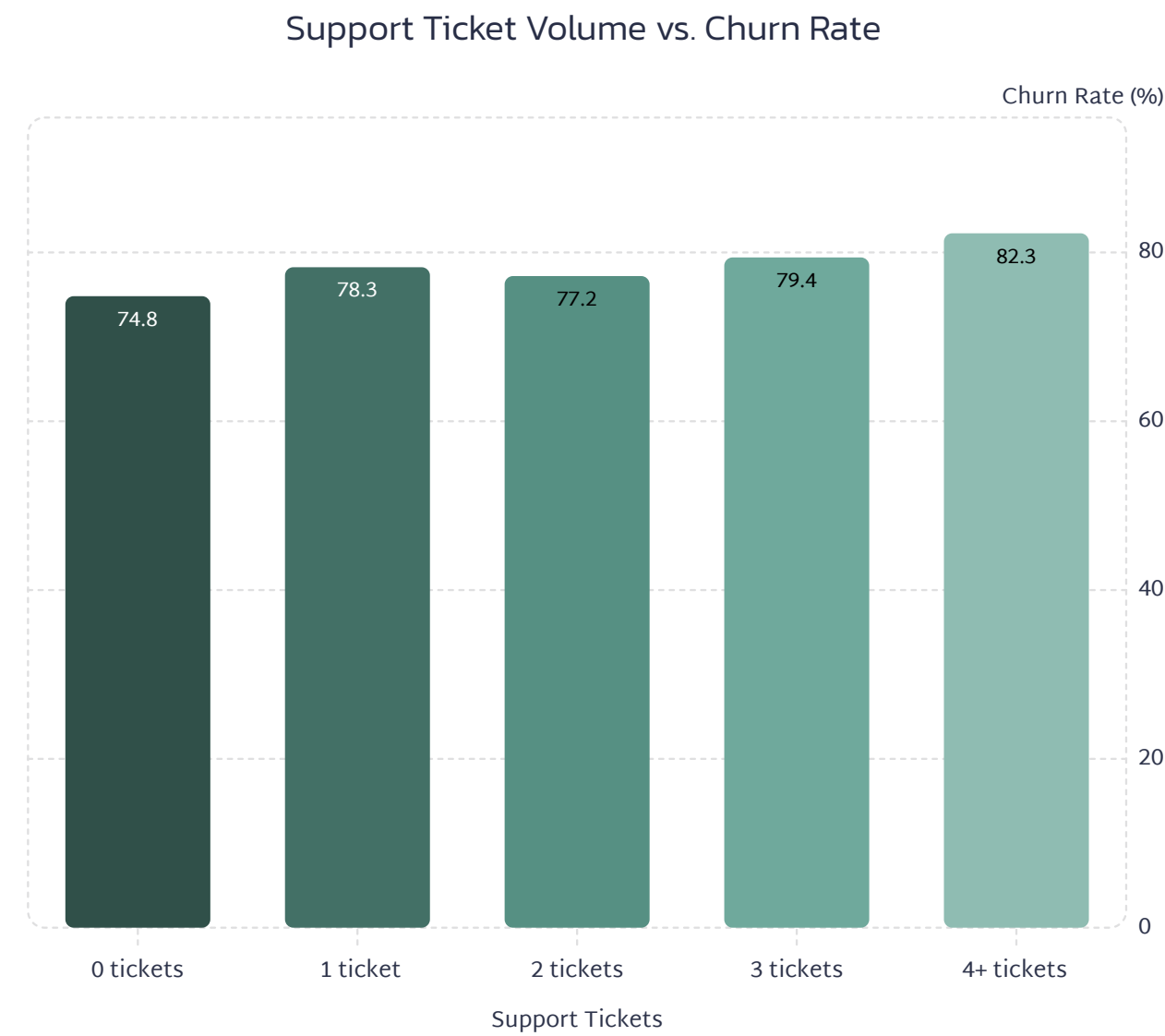
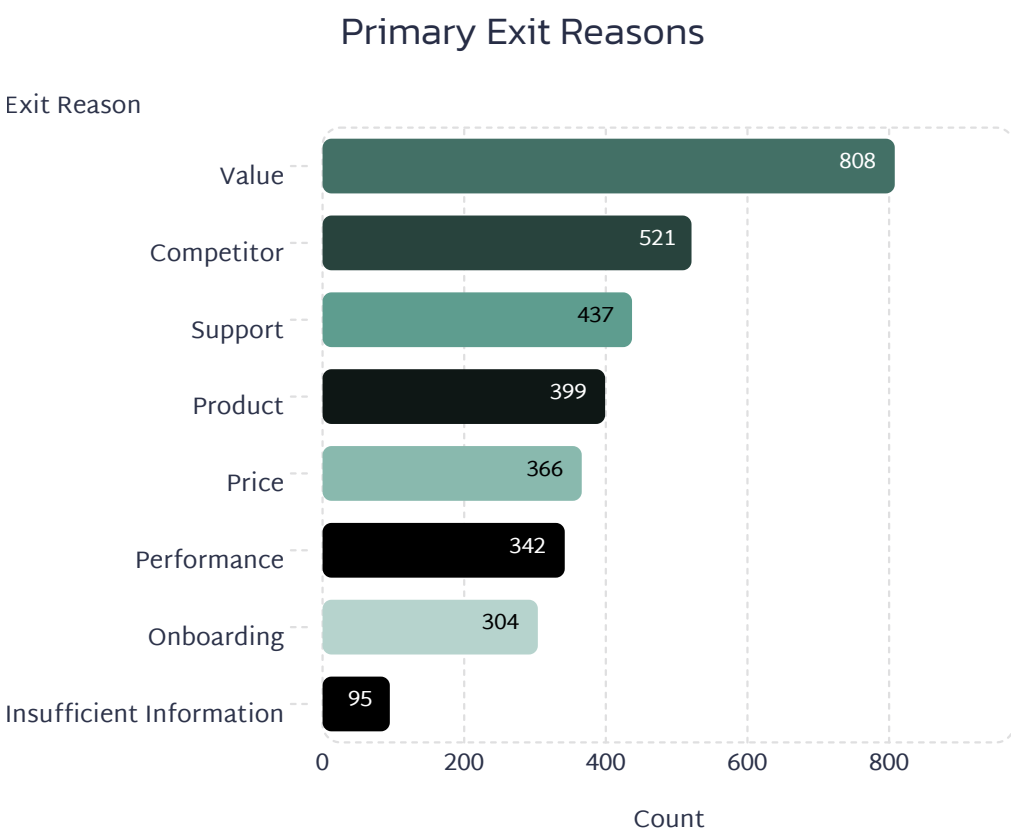
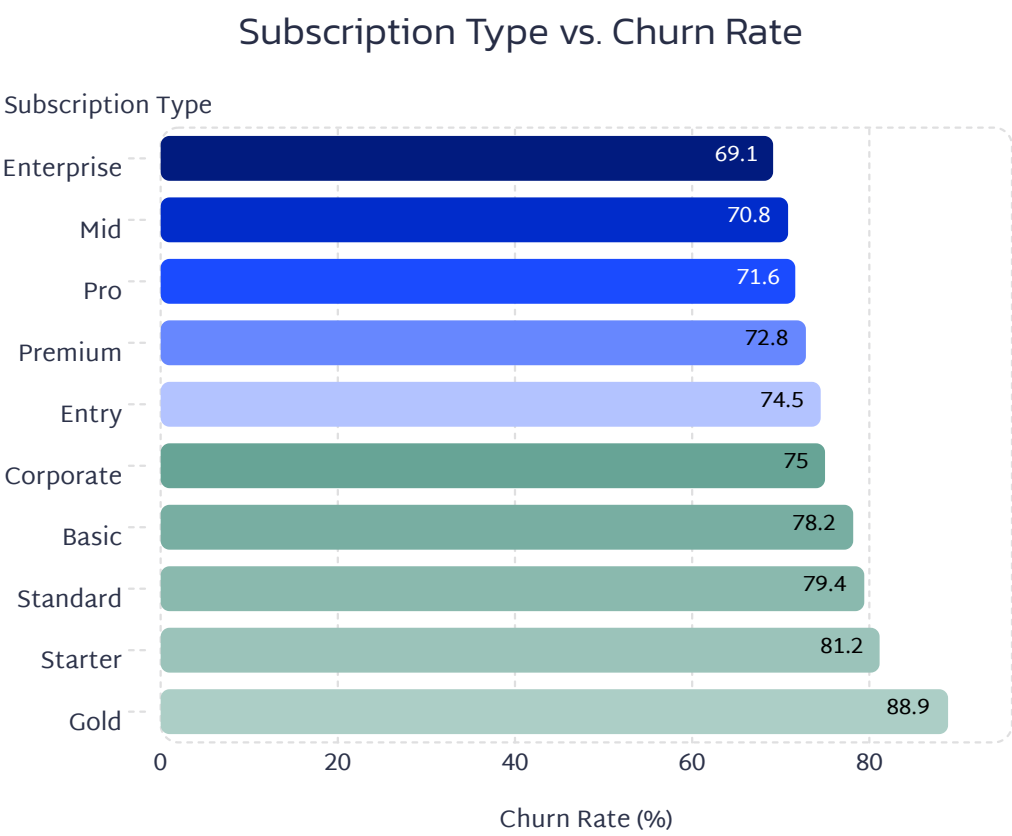
📄 **Prepared by:** Harpreet Singh

Tools & Technologies: Python •
Pandas • NumPy • NLP • SQL • Excel
• Power BI



Key Churn & Customer Insights

The integrated analysis examined 3,244 unique customers represented in 3,272 matched exit interviews. At the customer level, 2,506 customers had churned and 738 were retained, resulting in an observed churn rate of 77.25%. The analysis identified different observed churn patterns across customer segments, support activity, spending, subscription types, and exit-interview feedback.



Bottom takeaway: 77.25% observed customer churn | 2,506 churned customers | Value = largest exit category | Frustrated = dominant mood

 Note: Observed associations and patterns; not proof of causation.

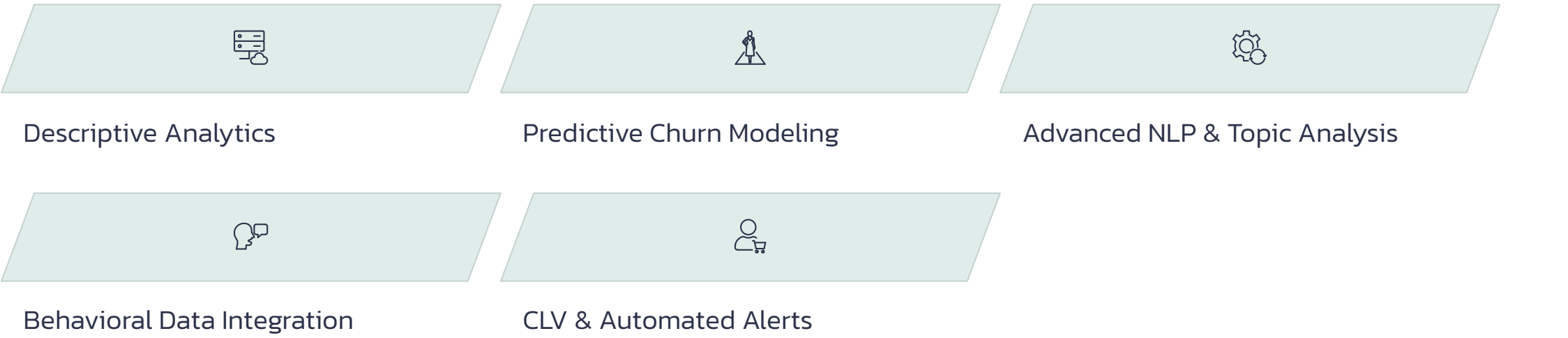
From Churn Insights to Action

The analysis translates customer and exit-interview patterns into practical areas for retention monitoring while establishing a foundation for future predictive analytics.

Immediate Actions:

- 1. Segment-Level Monitoring Monitor churn by subscription type, region, and other customer segments using multidimensional analysis.
- 2. Support Ticket Investigation Investigate recurring customer issues associated with higher support-ticket activity, particularly the 4+ ticket segment.
- 3. NLP Feedback Tracking Use NLP to continuously monitor exit reasons and customer mood to identify recurring customer-experience issues.
- 4. Dashboard Operationalization Use the Power BI dashboard for recurring monitoring as new customer and interview data become available.

Future Analytics Roadmap:



Future scope includes:

- Predictive churn probability
- Advanced NLP and topic modeling
- Product-usage and behavioral signals
- Customer Lifetime Value analysis
- Automated churn monitoring and alerts

Closing statement: From understanding customer churn to building a foundation for predictive retention.